Performing block-level recovery with Oracle Recovery Manager (RMAN)

SnapManager Oracle

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Performing block-level recovery with Oracle Recovery Manager (RMAN)

You can configure SnapManager to catalog its backups in Recovery Manager (RMAN), an Oracle tool, so that you can perform a block-level recovery using RMAN. RMAN can use either the database's control files or a separate recovery catalog database as its repository.

1. To perform a full offline backup using SnapManager, enter the following command:

   smo backup create -offline-full-profileprofile_name-labelbackup_label_name-verbose

   Where:

   - profile_name is the name of the profile associated with the backup
   - backup_label_name is the name of the backup label

```
smo backup create -offline -full -profile profile_monthly
-label full_backup -verbose

+
SMO-07109 [INFO ]: Cataloguing all files in backup set with RMAN
TAG=SMC_full_backup_1158773581857, RMAN=ES0/controlfile.
...
SMO-13037 [INFO ]: Successfully completed operation: Backup
SMO-13048 [INFO ]: Operation Status: SUCCESS
SMO-13049 [INFO ]: Elapsed Time: 0:02:20.506
Operation Id [ff8080810dcc47e3010dcc47eb7a0001] succeeded.
+
```

1. To verify that the backup is cataloged with RMAN, from the database host, enter the following command at the RMAN prompt:

   list datafilecopy tag tag_name;

```
RMAN> list datafilecopy tag SMO_full_backup_1158773581857;

Recovery Manager: Release 10.2.0.1.0 - Production on Wed Sep 20 10:33:41
2008
Copyright (c) 1982, 2008, Oracle. All rights reserved.
using target database control file instead of recovery catalog
 List of Datafile Copies
 Key File S Completion Time Ckp SCN Ckp Time Name
 ------- ---- - --------------- ---------- --------------- ---
 335 1 A 20-SEP-08 1347825 20-SEP-08
 /opt/<path>/smo/mnt/Host4_ES0_SMO_E_ES0_F_C_0_ff8080810dcc47e3010dcc47e
b7a0001
 /system01.dbf
 336 2 A 20-SEP-08 1347825 20-SEP-08
 /opt/<path>/smo/mnt/Host4_ES0_SMO_E_ES0_F_C_0_ff8080810dcc47e3010dcc47e
b7a0001
 /undotbs01.dbf
 334 3 A 20-SEP-08 1347825 20-SEP-08
 /opt/<path>/smo/mnt/Host4_ES0_SMO_E_ES0_F_C_0_ff8080810dcc47e3010dcc47e
b7a0001
 /sysaux01.dbf
 333 4 A 20-SEP-08 1347825 20-SEP-08
 /opt/<path>/smo/mnt/Host4_ES0_SMO_E_ES0_F_C_0_ff8080810dcc47e3010dcc47e
b7a0001
 /user01.dbf
 337 5 A 20-SEP-08 1347825 20-SEP-08
 RMAN>
```

2. To verify the database and determine if any blocks are corrupted, enter the following command:

   dbv FILE=user01.dbf

   The following output shows that two pages are corrupt:

```
DBVERIFY: Release 10.2.0.1.0 - Production on Wed Sep 20 13:35:44 2006
Copyright (c) 1982, 2005, Oracle. All rights reserved.
DBVERIFY - Verification starting : FILE = user01.dbf
Page 625 is marked corrupt
Corrupt block relative dba: 0x01400271 (file 5, block 625)
Bad header found during dbv:
Data in bad block:
type: 240 format: 6 rdba: 0xed323b81
last change scn: 0x6f07.faa74628 seq: 0x87 flg: 0x02
spare1: 0x60 spare2: 0x5 spare3: 0xef7d
consistency value in tail: 0xa210fe71
check value in block header: 0x13c7
block checksum disabled...
Page 627 is marked corrupt
Corrupt block relative dba: 0x01400273 (file 5, block 627)
Bad header found during dbv:
Data in bad block:
type: 158 format: 7 rdba: 0x2101e16d
last change scn: 0xe828.42414628 seq: 0xb4 flg: 0xff
spare1: 0xcc spare2: 0x81 spare3: 0x8665
consistency value in tail: 0x46d20601
check value in block header: 0x1a84
computed block checksum: 0x6c30
DBVERIFY - Verification complete
Total Pages Examined : 1280
Total Pages Processed (Data) : 1123
Total Pages Failing (Data) : 0
Total Pages Processed (Index): 0
Total Pages Failing (Index): 0
Total Pages Processed (Other): 34
Total Pages Processed (Seg) : 0
Total Pages Failing (Seg) : 0
Total Pages Empty : 120
Total Pages Marked Corrupt: 2
Total Pages Influx : 0
Highest block SCN : 1337349 (0.1337349)
```

3. To make the files from the backup accessible on the host and to RMAN, mount the backup by using the following command:

   smo backup mount -profileprofile_name-labellabel-verbose

```
smo backup mount -profile SALES1 -label full_backup -verbose

SMO-13046 [INFO ]: Operation GUID 8abc013111b9088e0111b908a7560001
starting on Profile SALES1
SMO-08052 [INFO ]: Beginning to connect mount(s) [/mnt/ssys1/logs,
/mnt/ssys1/data] from logical snapshot
SMO_SALES1_hsdb1_F_C_1_8abc013111a450480111a45066210001.
SMO-08025 [INFO ]: Beginning to connect mount /mnt/ssys1/logs from
snapshot SMO_SALES1_hsdb1_F_C_1_8abc013111a450480111a45066210001_0 of
volume hs_logs.
SMO-08027 [INFO ]: Finished connecting mount /mnt/ssys1/logs from
snapshot SMO_SALES1_hsdb1_F_C_1_8abc013111a450480111a45066210001_0 of
volume hs_logs.
SMO-08025 [INFO ]: Beginning to connect mount /mnt/ssys1/data from
snapshot SMO_SALES1_hsdb1_F_C_1_8abc013111a450480111a45066210001_0 of
volume hs_data.
SMO-08027 [INFO ]: Finished connecting mount /mnt/ssys1/data from
snapshot SMO_SALES1_hsdb1_F_C_1_8abc013111a450480111a45066210001_0 of
volume hs_data.
SMO-08053 [INFO ]: Finished connecting mount(s) [/mnt/ssys1/logs,
/mnt/ssys1/data] from logical snapshot
SMO_SALES1_hsdb1_F_C_1_8abc013111a450480111a45066210001.
SMO-13037 [INFO ]: Successfully completed operation: Backup Mount
SMO-13048 [INFO ]: Operation Status: SUCCESS
SMO-13049 [INFO ]: Elapsed Time: 0:01:00.981
Operation Id [8abc013111b9088e0111b908a7560001] succeeded.
```

4. To recover the blocks, in RMAN, enter the following command:

   blockrecover datafile ‘/mountpoint/path/file.dbf’ block block_id, from tag backup_rman_tag

```
RMAN> blockrecover datafile
'/mnt/ssys1/Host4_ES0/file01.dbf' block 625, 626, 627
from tag SMO_full_backup_1158773581857;

Starting blockrecover at 20-SEP-08 using target database control file
instead of recovery catalog
allocated channel: ORA_DISK_1
channel ORA_DISK_1: sid=153 devtype=DISK
channel ORA_DISK_1: restoring block(s) from datafile copy
 /opt/NetApp/smo/mnt/_mnt_ssys1_Host4_ES0_SMO_E_ES0_F_C_0_ff8080810dcc47
e3010dcc47eb7a0001/user01.dbf
starting media recovery
media recovery complete, elapsed time: 00:00:01
Finished blockrecover at 20-SEP-08
```

5. To verify if the blocks have been repaired, use the following command:

   dbv FILE=filename.dbf

   The following output shows that no pages are corrupt:

```
dbv FILE=user01.dbf

DBVERIFY: Release 10.2.0.1.0 - Production on Wed Sep 20 13:40:01 2008
Copyright (c) 1982, 2008, Oracle. All rights reserved.
DBVERIFY - Verification starting : FILE = user01.dbf
DBVERIFY - Verification complete
Total Pages Examined : 1280
Total Pages Processed (Data) : 1126
Total Pages Failing (Data) : 0
Total Pages Processed (Index): 0
Total Pages Failing (Index): 0
Total Pages Processed (Other): 34
Total Pages Processed (Seg) : 0
Total Pages Failing (Seg) : 0
Total Pages Empty : 120
Total Pages Marked Corrupt : 0
Total Pages Influx : 0
Highest block SCN : 1337349 (0.1337349)
```

   All corrupted blocks were repaired and restored.

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